HIGH-THROUGHPUT SEQUENCING ASSIGNMENT

RNA-Sequencing Differential Expression Analysis

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ABSTRACT

This report presents a complete RNA-sequencing analysis of the human colorectal carcinoma cell line HT55 exposed to itraconazole, a repurposed antifungal triazole under investigation as an anticancer agent, relative to DMSO-treated controls. Sequencing data were retrieved from SRA study SRP144496 (BioProject PRJNA454881), comprising eight single-end Illumina runs corresponding to four biological replicates per condition. Raw reads underwent quality assessment with FastQC and MultiQC, splice-aware alignment to the GRCh38 reference genome (GENCODE release 50 annotation) with HISAT2, gene-level quantification with featureCounts, and differential expression modelling with DESeq2 in R using a negative-binomial generalized linear model and Benjamini-Hochberg false discovery rate control.

Of 78,941 annotated GENCODE genes, 14,492 passed an independent expression filter (raw count ≥ 10 in at least four samples) and were carried forward into statistical testing. At an adjusted p-value threshold of 0.05 and an absolute log2 fold-change threshold of 1.0, 1,570 genes were differentially expressed between itraconazole-treated and control HT55 cells, comprising 845 upregulated and 725 downregulated transcripts. The most significantly upregulated genes included ALDOC, BMF, ACSL1, PAQR5 and SLC6A20, while the most significantly downregulated genes included FGF19, MYH7B, CDCA7, LINC02418 and MAP1B. These changes are consistent with a transcriptional programme combining metabolic reprogramming, induction of pro-apoptotic signaling, and suppression of proliferative growth-factor signaling, in line with the reported pharmacological activity of itraconazole in colorectal cancer models. Principal component analysis across the full sixteen-sample dataset showed that cell line of origin, not treatment, explained the majority of transcriptomic variance (83%), with treatment status resolved on a secondary axis (13%), underscoring the importance of the paired, cell-line-stratified contrast used in this analysis.

1.0 INTRODUCTION

1.1 Study Background

SRP144496 is a publicly deposited RNA-sequencing study (BioProject PRJNA454881, Homo sapiens) profiling the transcriptional effects of itraconazole exposure in two colorectal carcinoma cell lines, HT55 and SW948. Itraconazole is a triazole antifungal agent that has attracted interest as a repurposed oncology compound on the basis of reported activity against Hedgehog/Smoothened signaling, angiogenesis, cholesterol trafficking, and mTOR pathway components, several of which are implicated in colorectal tumour growth and maintenance. Characterizing the transcriptomic response to itraconazole in colorectal cancer cell lines provides a basis for generating mechanistic hypotheses and candidate biomarkers of drug response ahead of functional validation.

This report addresses the HT55 cell line specifically, comparing itraconazole-treated cells against DMSO vehicle controls. HT55 is a moderately differentiated colorectal adenocarcinoma line frequently used in pharmacogenomic and drug-repurposing studies. The parallel SW948 comparison present in the same study is referenced only where relevant for interpretive context in the Discussion, and is not the primary subject of this analysis.

1.2 Sample Metadata

Sixteen single-end Illumina runs (50 bp reads) are associated with SRP144496 in total, split evenly between the HT55 and SW948 cell lines and, within each cell line, evenly between itraconazole and control conditions (four biological replicates per group). The eight runs analyzed in this report, corresponding to the HT55 cell line, are summarized below.

Table 1. HT55 sample metadata for SRP144496

Run Accession	Sample Title	Cell Line	Condition	Replicate	Library Layout
SRR7108388	HT55_CONT_1	HT55	Control (DMSO)	1	Single-end
SRR7108389	HT55_CONT_2	HT55	Control (DMSO)	2	Single-end
SRR7108390	HT55_CONT_3	HT55	Control (DMSO)	3	Single-end
SRR7108391	HT55_CONT_4	HT55	Control (DMSO)	4	Single-end
SRR7108392	HT55_ITRA_1	HT55	Itraconazole	1	Single-end
SRR7108393	HT55_ITRA_2	HT55	Itraconazole	2	Single-end
SRR7108394	HT55_ITRA_3	HT55	Itraconazole	3	Single-end
SRR7108395	HT55_ITRA_4	HT55	Itraconazole	4	Single-end

All sixteen runs (HT55 and SW948 combined) were processed through an identical pipeline to preserve comparability. Only the eight HT55 runs and the corresponding itraconazole-versus-control contrast are reported here in detail.

2.0 METHODS

2.1 Raw Read Quality Assessment and Pre-processing

Raw FASTQ files were retrieved and integrity-checked against expected file sizes and gzip checksums prior to analysis. Per-sample quality assessment was performed with FastQC, and results were aggregated across all sixteen runs with MultiQC. No adapter or quality trimming step was applied prior to alignment. The FastQC Adapter Content module passed for all samples, indicating negligible adapter read-through at the assessed read positions, and the splice-aware aligner used downstream tolerates soft-clipping of low-quality read ends.

2.2 Reference Genome Indexing and Alignment

Reads were aligned to the GRCh38 primary assembly using the GENCODE release 50 annotation with HISAT2, a splice-aware short-read aligner. A HISAT2 index incorporating known splice junctions (extracted from the GENCODE GTF with hisat2_extract_splice_sites.py) was built once and reused across all samples. Each sample was aligned independently in single-end mode. Output alignments were coordinate-sorted and indexed with SAMtools, and per-sample mapping statistics were collected with samtools flagstat and the native HISAT2 alignment summary, both aggregated with MultiQC.

2.3 Expression Quantification

Gene-level read counts were generated with featureCounts (Subread package) using exon-level summarization to the gene_id attribute of the GENCODE v50 GTF, under a reverse-stranded library assumption (featureCounts strand mode 2), consistent with strandedness empirically confirmed for the library preparation protocol prior to final quantification. All sixteen sorted BAM files were quantified in a single featureCounts invocation to produce one integer count matrix (78,941 GENCODE gene records × 16 samples), which was reformatted into a clean, DESeq2-ready count matrix, alongside per-sample read-assignment statistics.

2.4 Statistical Modelling and Differential Expression Testing

Differential expression was modelled in R with DESeq2, using a single composite design factor encoding the combination of cell line and treatment ("group"), which allows independent, cell-line-stratified pairwise contrasts to be extracted from one fitted model while sharing dispersion information across all sixteen samples. Prior to model fitting, an independent low-expression filter retained genes with a raw count of at least 10 in at least 4 of the 16 samples (78,941 genes reduced to 14,492). Size factors were estimated by the median-of-ratios method, gene-wise dispersions were shrunk toward a fitted mean-dispersion trend, and significance was assessed per gene with a Wald test. Genes were classified as significantly differentially expressed at a Benjamini-Hochberg adjusted p-value (FDR) below 0.05 and an absolute log2 fold-change above 1.0. Resulting gene identifiers were annotated with gene symbol and biotype using the matching GENCODE v50 gene table.

Table 2. DESeq2 filtering and differential expression testing parameters applied to all pairwise contrasts

Parameter	Value
Low-expression filter	raw count $\geq$ 10 in $\geq$ 4 / 16 samples
Genes before filtering	78,941
Genes after filtering	14,492
Normalisation	DESeq2 median-of-ratios size factors
Dispersion model	Negative-binomial GLM, empirical Bayes shrinkage
Test statistic	Wald test per contrast
Significance (FDR)	Benjamini-Hochberg adjusted p-value < 0.05
Effect-size threshold	|log2 fold-change| > 1.0

3.0 RESULTS

3.1 Raw Read Quality (Pre-processing QC)

All eight HT55 libraries passed FastQC's Basic Statistics, Per-Sequence Quality Score, Per-Base N Content, Sequence Length Distribution and Adapter Content modules, with no reads flagged as poor quality. Mean base-call quality across read positions ranged from Phred 33 (3′ read end) to Phred 37 (read body), averaging approximately Phred 36.7 per sample. GC content was uniform across samples at 49%, consistent with the human transcriptome and free of the multi-modal distributions that typically flag contamination. Two modules were flagged in every sample: Per

Base Sequence Content (FAIL) and Sequence Duplication Levels (FAIL). Per-base sequence content failures of this kind are expected for RNA-seq libraries, which lack the uniform base composition of genomic DNA due to non-random fragmentation and priming bias at read starts, and duplication-level failures are similarly routine for cell-line RNA-seq, where a comparatively small number of highly expressed transcripts are sequenced to saturation; neither finding indicates a technical quality defect. Per Tile Sequence Quality returned a WARN for all samples, consistent with minor flow-cell positional effects that did not translate into reduced overall base quality.

3.2 Read Alignment Efficiency

HISAT2 achieved a high overall alignment rate across all HT55 samples, averaging 98.4% (range 97.4-98.7%), with samtools flagstat confirming 98.2–99.0% of alignment records mapped to the reference. These rates are consistent across replicates and treatment groups, indicating that library complexity and read quality were comparable between conditions and that alignment efficiency is not a confounder of the downstream differential expression contrast.

3.3 Feature Quantification Summary

featureCounts assigned a mean of 53.8% of aligned records to annotated exonic gene features across the eight HT55 libraries (range 51.1-55.3%), with the remainder attributable chiefly to intronic and intergenic alignments and reads overlapping multiple genes. Such assignment profile is typical of single-end, exon-level RNA-seq quantification without prior transcript-level deduplication. Assignment rates were consistent between control and itraconazole-treated samples, indicating that the featureCounts step introduced no systematic bias between conditions.

3.4 Sample Relationships: Principal Component Analysis and Correlation Structure

Principal component analysis of variance-stabilized expression values across the full sixteen-sample dataset showed that PC1, explaining 83% of total variance, separates samples by cell line of origin (HT55 versus SW948), while PC2, explaining 13% of variance, separates samples by treatment condition (itraconazole versus control) within each cell line. This structure confirms that cell-line identity is the dominant source of transcriptomic variation in the combined dataset and justifies the cell-line-stratified contrast design used for differential expression testing, rather than a pooled analysis across cell lines.

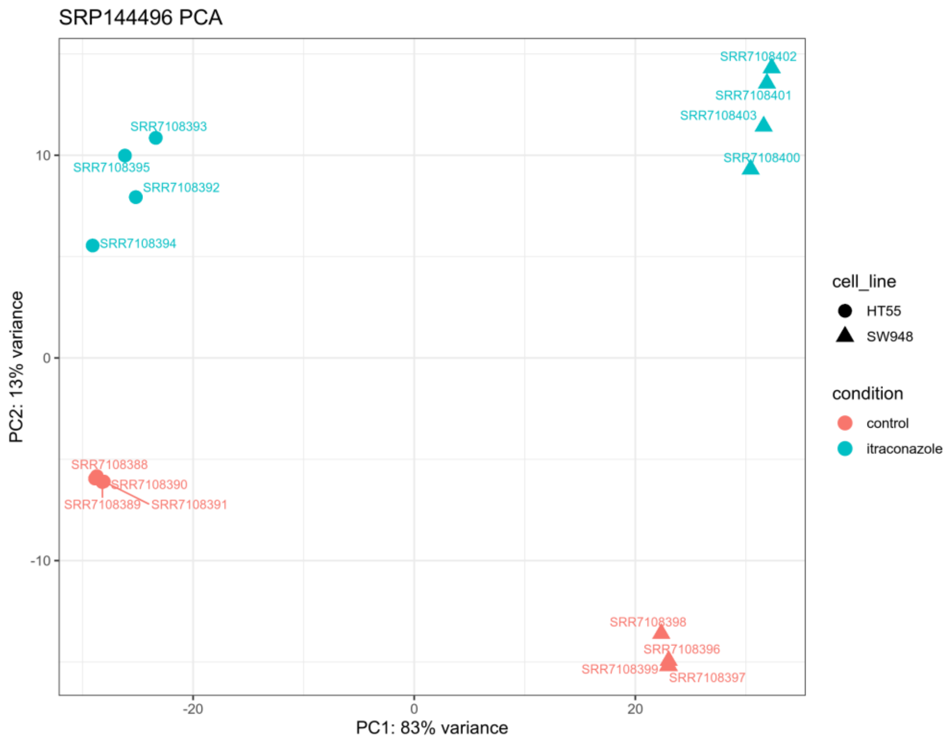

Figure 1. Principal component analysis (PC1 vs PC2) of variance-stabilised expression across all sixteen SRP144496 samples. Shape denotes cell line (HT55: circle, SW948: triangle); colour denotes treatment condition

Sample-to-sample correlation of normalized expression profiles recapitulates this structure: HT55 control and HT55 itraconazole replicates each form tight, internally consistent clusters, distinct from the corresponding SW948 clusters, with HT55 itraconazole samples showing clearly reduced correlation to HT55 controls relative to their within-group correlation.

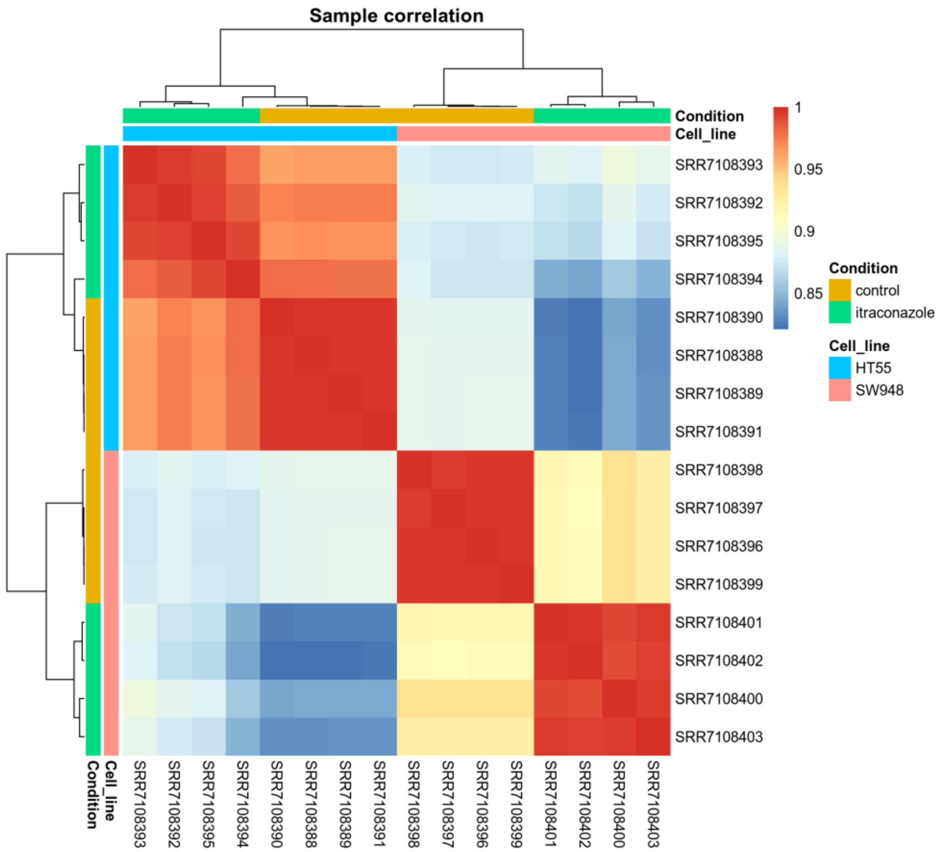

Figure 2. Pairwise Pearson sample correlation heatmap with hierarchical clustering, annotated by cell line and treatment condition, across all sixteen samples

3.5 Dispersion Estimation

The DESeq2 dispersion plot shows gene-wise dispersion estimates (black) shrunk toward the fitted mean-dispersion trend (red) to produce the final dispersion estimates used for testing (blue), with a small number of high-dispersion outlier genes retained without shrinkage (blue-circled points) as expected under DESeq2's outlier-handling procedure. The trend follows the expected inverse relationship between mean expression and dispersion, with no evidence of overdispersion pathology that would compromise the validity of the Wald tests performed downstream.

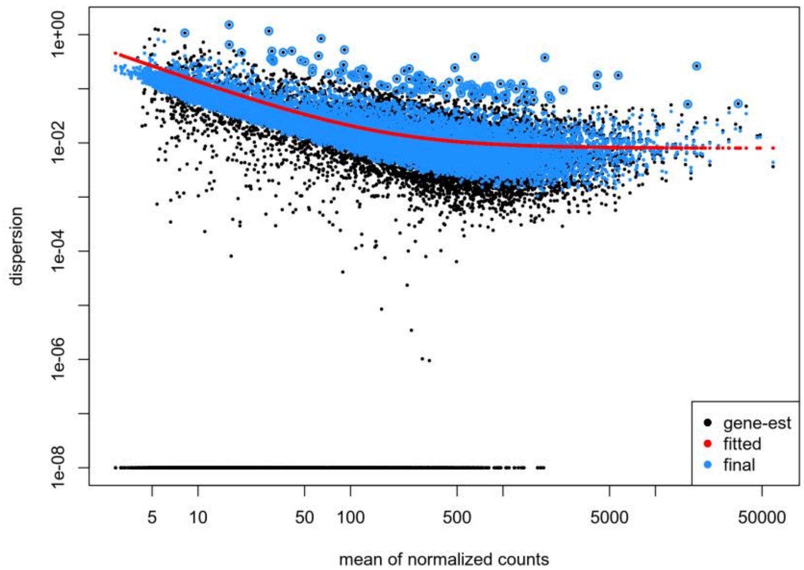

Figure 3. DESeq2 dispersion estimates versus mean normalized count, across the full sixteen-sample DESeq2 model fit

3.6 Differential Expression: HT55 Itraconazole versus Control

The HT55 itraconazole-versus-control contrast yielded 1,570 significant genes: 845 upregulated and 725 downregulated by itraconazole exposure. For comparative context, the parallel SW948 itraconazole-versus-control contrast in the same dataset yielded a substantially larger set of 4,731 significant genes (2,064 upregulated, 2,667 downregulated) under identical thresholds, indicating a more extensive transcriptional response in that cell line.

Table 3. Differential expression summary for the HT55 itraconazole-versus-control contrast (padj < 0.05, |log2FC| > 1.0)

Comparison	Upregulated	Downregulated	Total Significant
HT55 itraconazole vs. control	845	725	1,570

3.6.1 Volcano Plot

The volcano plot displays statistical significance (−log10 adjusted p-value) against effect size (log2 fold-change) for all 14,492 tested genes. The distribution is markedly asymmetric toward upregulation at the extremes of statistical significance. The single most significant gene, ALDOC, and several of the next most significant genes (BMF, ACSL1, PAQR5, SLC6A20) are all upregulated, whereas the most significant downregulated gene, FGF19, reaches a comparatively lower significance ceiling. This pattern indicates that itraconazole exposure induces a strong, highly reproducible upregulation signal in a defined subset of genes, superimposed on a broader, more diffuse pattern of both up- and downregulated genes closer to the significance threshold.

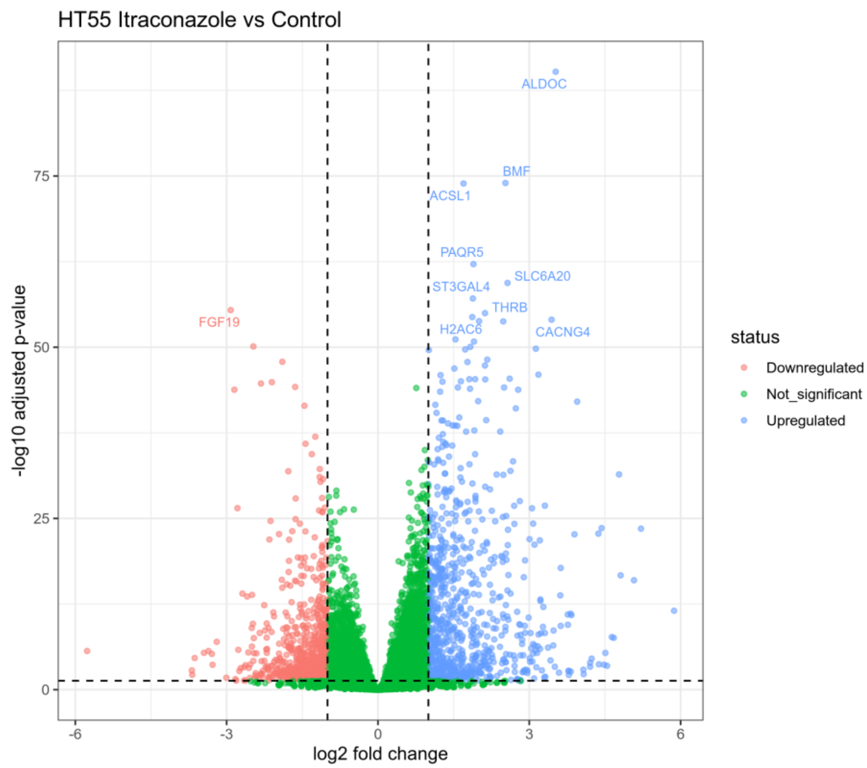

Figure 4. Volcano plot of the HT55 itraconazole-versus-control contrast. Dashed lines mark the padj = 0.05 and |log2FC| = 1.0 thresholds; the ten most significant genes are labelled

3.6.2 MA Plot

The MA plot (log fold-change against mean normalized count) shows significant genes (blue) distributed across the full range of expression levels, from lowly expressed transcripts (mean count ≈ 5) to highly expressed transcripts (mean count > 10,000), with fold-change magnitude compressing toward zero at high expression levels as expected under DESeq2's shrinkage-free MA visualization. The absence of a strong intensity-dependent bias (no systematic tilt of the point cloud) supports adequate normalization of the count data between conditions.

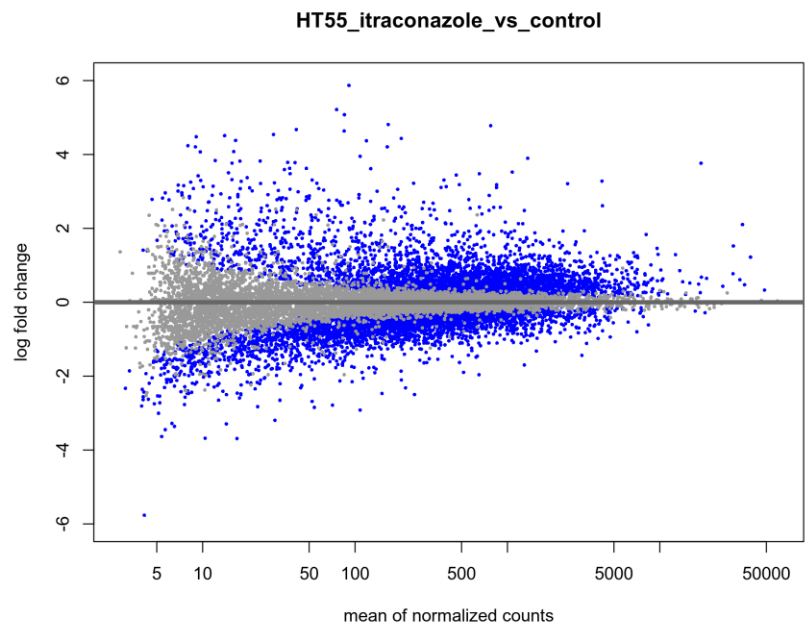

Figure 5. MA plot of the HT55 itraconazole-versus-control contrast. Blue points indicate genes meeting both significance thresholds; grey points are non-significant

3.6.3 Top Differentially Expressed Genes

The ten most statistically significant upregulated and downregulated genes (ranked by adjusted p-value) are listed below.

Table 4. Top ten upregulated and top ten downregulated genes in HT55 cells following itraconazole exposure, ranked by adjusted p-value. All genes listed are protein-coding except LINC02418 (long non-coding RNA)

Rank	Gene	log2FC	Adjusted p-value	Direction
1	ALDOC	+3.52	5.9e−91	Up
2	BMF	+2.53	1.0e−74	Up
3	ACSL1	+1.70	1.2e−74	Up
4	PAQR5	+1.89	7.3e−63	Up
5	SLC6A20	+2.57	4.1e−60	Up
6	ST3GAL4	+1.88	7.6e−58	Up
7	THRB	+2.12	1.1e−55	Up
8	H2AC6	+1.87	3.9e−55	Up
9	CACNG4	+3.44	9.5e−55	Up
10	CTSS	+2.00	1.6e−54	Up
1	FGF19	−2.92	3.8e−56	Down
2	MYH7B	−2.47	7.6e−51	Down
3	CDCA7	−1.90	1.3e−48	Down
4	LINC02418	−2.10	1.3e−45	Down

5	MAP1B	−2.32	2.0e−45	Down
6	SLC43A2	−1.64	6.3e−45	Down
7	NTSR1	−2.85	1.6e−44	Down
8	HMGN2	−1.46	3.5e−42	Down
9	TSEN15	−1.24	1.2e−37	Down
10	RBM3	−1.44	1.2e−36	Down

3.6.4 Top-50 DEG Heatmap

Unsupervised hierarchical clustering of the fifty most significant differentially expressed genes cleanly separates the four control replicates from the four itraconazole-treated replicates along the sample axis, with two well-defined gene clusters: a smaller block of genes suppressed by treatment (including FGF19, MAP1B, CDCA7 and HMGN2) and a larger block induced by treatment (including ALDOC, ACSL1, BMF, ST3GAL4 and CACNG4). Replicate concordance within each condition is high, with only modest within-group variation in relative expression intensity.

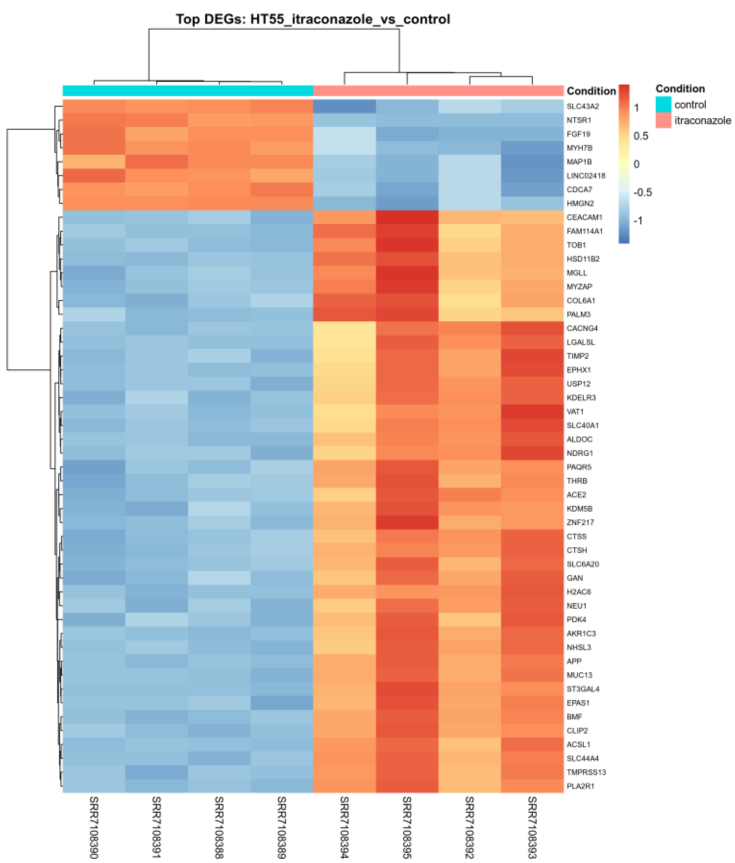

Figure 6. Hierarchically clustered heatmap of the top 50 differentially expressed genes (by adjusted p-value) in the HT55 itraconazole-versus-control contrast, with row-wise z-score scaling of variance-stabilised expression

4.0 Discussion

4.1 Interpretation of the Transcriptional Response

The differential expression profile observed in HT55 cells following itraconazole exposure is consistent with several previously reported mechanisms of itraconazole action in cancer cell models, though the present analysis is transcriptomic and hypothesis-generating rather than confirmatory of any specific mechanism. Among the most significantly upregulated genes, BMF encodes a pro-apoptotic BH3-only protein, and its induction is consistent with engagement of intrinsic apoptotic signaling, a plausible contributor to itraconazole's reported cytotoxic activity in colorectal cancer lines. ALDOC (aldolase C) and ACSL1 (acyl-CoA synthetase long-chain family member 1) are both central to glycolytic and lipid metabolism, respectively. Their coordinate upregulation suggests a metabolic reprogramming response, which is compatible with itraconazole's established interference with cholesterol trafficking and sterol biosynthesis in treated cells.

Among downregulated genes, FGF19 is a growth factor implicated in colorectal tumour proliferation through FGFR4-dependent signaling, and its suppression is consistent with reduced mitogenic signaling under treatment. CDCA7 (cell division cycle associated 7) and HMGN2, both associated with active cell-cycle progression and chromatin accessibility in proliferating cells, were also significantly downregulated, in line with a treatment-induced reduction in proliferative capacity. In summary, the pattern of induced apoptotic and metabolic-stress genes alongside suppressed proliferation- and growth-factor-associated genes is consistent with a cytostatic-to-cytotoxic transcriptional programme, though direct functional validation (for example, apoptosis assays, cell-cycle profiling, or FGF19/FGFR4 pathway reporter assays) would be required to confirm these mechanisms in HT55 cells specifically.

4.2 Comparison with SW948

The substantially larger number of significant DEGs detected in SW948 (4,731) relative to HT55 (1,570) under identical statistical thresholds indicates that itraconazole elicits a more extensive transcriptional response in SW948 cells, which may reflect differences in genetic background, baseline pathway dependency, or drug sensitivity between the two colorectal cancer lines. This asymmetry cannot be attributed to technical factors. Both cell lines were processed through an identical pipeline with comparable sequencing depth, alignment rate and quantification efficiency, and the composite DESeq2 design shares dispersion estimation across all sixteen samples. A systematic overlap analysis of the two significant DEG sets (shared versus cell-line-specific responses) would clarify whether the HT55 response represents a subset of a shared itraconazole signature or a distinct, cell-line-specific programme, and is a natural extension of the present analysis.

4.3 Limitations

Replicate size: four biological replicates per condition provide reasonable power for genes with large effect sizes but limited power to detect genes with modest fold-changes near the significance threshold, particularly given inter-replicate variability in library size (10.5-13.5 million raw reads).

No adapter/quality trimming: reads were aligned directly from raw FASTQ files without a dedicated trimming step. FastQC's Adapter Content module passed for all samples, mitigating this concern, but residual low-quality 3' bases (mean Phred ≈ 33 at the final read positions) were not explicitly removed prior to alignment, relying instead on the aligner's local/soft-clipping behaviour.

Single-end, single time-point design: 50 bp single-end reads limit isoform-level resolution relative to paired-end protocols, and the dataset captures a single post-treatment time point, precluding inference of response kinetics.

In vitro, single-line generalizability: findings are derived from an immortalized cell line under defined culture conditions and may not generalize directly to primary tumour tissue or in vivo drug exposure, where pharmacokinetics, tumour microenvironment and stromal interactions differ substantially.

Batch/technical covariates: the sample metadata available for this dataset does not encode sequencing batch or lane information separately from biological replicate, so any residual batch effect cannot be explicitly modelled or ruled out, although the tight within-group clustering observed in Figures 1 and 2 argues against a dominant batch confound.

Annotation vintage: gene models are based on GENCODE release 50; results may shift marginally under alternative or more recent annotation builds, particularly for non-coding and low-confidence transcripts.